DEBOPAM GHOSH

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Codeforces | LinkedIn | GitHub | LeetCode

Professional Summary

Final-year Computer Science and Business Systems undergraduate with hands-on experience in backend and systems-oriented development in C, C++, Java, and Python, including operating systems, computer networks, and DBMS coursework. Skilled at debugging and validating system behavior through structured testing, confidence-threshold data matching, and uptime/health monitoring across production backend services. Proven ability to design reliable, scalable REST APIs, shown through CI/CD health-check monitoring achieving 99% uptime and an AI-assisted matching engine with an 80%+ confidence threshold. National Finalist, Smart India Hackathon 2024, and Top 16% globally in Amazon ML Challenge 2025.

Education

Academy of Technology, Kolkata, WB *2023 – 2027(June)*
Bachelor of Technology in Computer Science and Business Systems (GPA: 7.69)

Technical Skills

Languages: Java (11+), C, C++, Python, JavaScript
Backend Frameworks: Spring Boot, Node.js, Express.js, Flask, FastAPI
Frontend & ML: React.js, TensorFlow, Keras, Scikit-learn, Pandas, NumPy
Databases: MySQL, MongoDB, MongoDB Atlas, Supabase (RDBMS and NoSQL), PostgreSQL
Tools: Git, GitHub, Docker, Vercel, Render, VS Code
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems, Machine Learning, Testing and Code Quality

Projects

Sanvix | *FastAPI, Supabase (PostgreSQL, Auth, RLS), Python, React 19* [GitHub]

- Engineered a FastAPI backend with 8+ REST endpoints (authentication, medicine search, prescription scanning, subscriptions, payments) backed by Supabase PostgreSQL, as backend lead in a 5-member hackathon team, following clean code and maintainability standards.
- Built and validated an AI-assisted prescription-scanning pipeline using fuzzy string matching to resolve OCR-extracted, misspelled medicine names, tuning and testing a confidence-threshold match (>80%) to ensure reliable output, plus an automated refill-reminder engine.
- Secured multi-patient access with Supabase Auth and Row-Level Security; deployed and independently tested the backend on Render, decoupled from the Vercel-hosted React/TypeScript frontend, to isolate and debug service-level issues.

Code Arena | *Node.js, Express.js, REST APIs, Socket.io, MongoDB Atlas, React.js* [GitHub]

- Designed RESTful backend APIs using Node.js and Express.js to manage 300+ algorithm challenges with automated test-case validation and difficulty scaling, alongside a secure sandboxed remote code execution engine for C++/Python submissions.
- Engineered a low-latency real-time collaboration engine using Socket.io, achieving sub-100ms synchronization through iterative performance testing and tuning.
- Deployed on Render with 99% uptime, implementing automated CI/CD health-check cron jobs to continuously monitor system reliability and surface failures early.

AssetPulse | *Java 21, Spring Boot, Spring Data JPA, PostgreSQL, REST API, Maven, Lombok* [GitHub]

- Built a production-ready REST API using Spring Boot and Spring Data JPA to manage assets (create, retrieve, list, delete) with a PostgreSQL backend and UUID-based entity identification.
- Designed a layered backend architecture (Controller → Service → Repository) following SOLID principles, with robust exception handling to improve fault tolerance and code maintainability.
- Packaged the application as a runnable JAR via Maven with unit test coverage using JUnit, targeting Java 21 with a configurable datasource for PostgreSQL or in-memory database environments to support consistent testing.

Certifications

Machine Learning Specialization – Issued by Andrew Ng via Coursera

Achievements and Leadership

- Smart India Hackathon 2024: Institutional Finalist, selected and nominated by Academy of Technology among top teams for National Finals consideration.
- Amazon ML Challenge 2025: Ranked 1309 out of 8000+ (Top 16%) globally, reflecting strong problem-solving, analytical, and logical reasoning skills.
- Competitive Programming: Codeforces participant (Max Rating: 1223), consistent practice in data structures, algorithms, and debugging logic-heavy solutions.